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November 7, 2026 · **Subject: Submission of Original Research Manuscript for Double-Blind Review**

Editor-in-Chief, *Journal of Imaging Informatics in Medicine (JIIM)*, Society for Imaging Informatics in Medicine (SIIM), Springer Nature

Dear Editor-in-Chief,

We are pleased to submit our original research manuscript entitled **"Disentangling Evaluation Distortions in 2D-to-3D Brain Tumor MRI Segmentation: A Patient-Grouped Leakage, Calibration, and Volumetric Post-Processing Audit on TCGA-LGG"** for consideration as an Original Research Article in the *Journal of Imaging Informatics in Medicine* (JIIM).

Synergy with JIIM's Mission and Recent Auditing Literature: Addressing JIIM's core mission to advance imaging informatics rigor, AI safety, and evaluation reproducibility, our work complements and extends the critical benchmarking audit recently published in JIIM by Uddin and Ferdouse (Nov 2026, DOI: 10.1007/s10278-026-02489-8) from 2D classification to volumetric MRI segmentation. While Uddin and Ferdouse demonstrated that image-level partitioning inflates 2D brain tumor classification accuracy from 91.82% to 97.10% on the benchmark subset providing patient identifiers, our study provides a systematic, multi-layer evaluation audit on the multi-site TCGA-LGG benchmark (110 patients, 3,929 axial slices, five institutions).

Key Findings and Methodological Contributions:

- **Empirical Leakage, Appearance Probe, & Mask Continuity Index (MCI):** Random-slice partitioning artificially inflates positive-slice Dice by $+\mathbf{8.79}$ pp (73.96% to 82.74%, 95% CI: $[+6.16, +11.99]$ pp) and all-slice Dice by $+\mathbf{6.20}$ pp. A parameter-free copy baseline reaches 86.18% all-slice Dice via empty-slice copying ($\mathrm{FPR}_{\mathrm{empty}} = 0.63\%$). Per-patient network inflation shows no detectable linear association with mask continuity (MCI $r = -0.068$, TOST $p = 0.0263$ within $|\Delta r| < 0.25$) and persists across non-adjacent slices (primarily at a 2-slice gap, $\Delta z \geq 2$: matched $+5.54$ pp). A non-lesional brain texture probe predicts exact patient identity with 79.64% accuracy (82.32% at $\Delta z = 1$, 59.44% at $\Delta z \geq 2$; vs. 0.91% chance), indicating high patient-identifying information in non-lesional slices, although neither mask continuity nor background similarity linearly predicts per-patient inflation magnitude.
- **Analytical Metric Dilution, Critical Threshold ($\pi^* = 86.53\%$), & Rank Decorrelation:** We analytically formulate empty-slice dilution and derive the critical rank-inversion threshold $\pi^* = 86.53\%$ [95% CI: $75.56\%, 99.73\%$]. Sweeping prevalence reveals no detectable rank agreement (Kendall $\tau = -0.067$, 95% CI: $[-0.156, +0.111]$) and top-rank reversal, shifting rankings between global false-positive suppression (Largest-CC) and boundary-recovery rules. A 2^3 factorial attribution decomposes total $+18.14$ pp inflation into dilution (45.7%), leakage (40.8%), and post-processing (13.5%).
- **Calibration Invariance & Conformal Coverage Degradation:** We show analytically that monotonic calibration preserves validation-optimized spatial overlap (Remark 1). Weighted active-bin calibration error decreases from 36.14% to 29.57% under single-parameter temperature scaling and to 8.46% under two-parameter affine Platt scaling. In a single-model conformal audit, calibrating on in-sample training-patient slices reduces test coverage to 76.96% (median 85.87%, mean drop -8.78 pp), whereas honest patient calibration achieves 85.74% ($90.08\% \pm 1.86\%$ with $n_{\mathrm{cal}} = 30$, and 87.69% under patient-level CRC).
- **Comparative 3D Post-Processing Audit (CAL-3DH):** Balanced component filtering and localized apical recovery ($+0.28$ pp across 17 volumes) without surface distance penalty, with reconstructed volumes demonstrating good volumetric consistency ($\mathrm{ICC}(2,1) = 0.8894$, with wide individual limits of agreement) and exploratory molecular-cluster separation between IDH-mutant non-codel COC 1 and IDH-wildtype-like COC 2 ($p = 0.0268$, covariate-adjusted $p = 0.056$).

Compliance Declarations: The manuscript strictly adheres to double-blind guidelines (`manuscript_blinded.pdf` contains zero identifying metadata). Complete author details, ethics approvals, and declarations are provided on `title_page.pdf`. All source code, reproduction scripts, and 5-fold cross-validation pipelines are provided in an anonymized repository for peer review and will be made openly available on GitHub upon acceptance (with full repository details provided on `title_page.pdf`). All 8 vector figures are prepared in flat directory format (`Fig1.pdf`–`Fig8.pdf`). This manuscript has not been published and is not under consideration elsewhere. All authors have approved submission.

Sincerely yours,

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